EXPERIMENT 10 C : DESIGN AND CONFIGURATION OF AN INTERNETWORK IN THE LAB USING SWITCH, ROUTER AND ETHERNET CABLES

AIM:

To design and configure a basic internetwork using switches, a router and Ethernet cables, assign IP addresses to end devices, and verify communication within and between networks.

REQUIREMENTS:

Two PCs, two switches, one router, Ethernet straight-through cables and a computer running Cisco Packet Tracer (or equivalent lab equipment).

PROCEDURE:

1. Open Cisco Packet Tracer and create a new project.
2. Place two PCs, two switches and one router in the workspace.
3. Connect PC0 to Switch0 and PC1 to Switch1 using Ethernet straight-through cables.
4. Connect each switch to a separate router interface. Use the interface names available on the selected router model.
5. Configure PC0 with IP address 192.168.10.10, subnet mask 255.255.255.0 and default gateway 192.168.10.1.
6. Configure PC1 with IP address 192.168.20.10, subnet mask 255.255.255.0 and default gateway 192.168.20.1.
7. Configure the router interfaces with 192.168.10.1/24 and 192.168.20.1/24 respectively, then enable them using no shutdown.
8. Wait for the switch and router links to become active. Verify interface status using show ip interface brief.
9. From PC0, ping its default gateway and then ping PC1. Repeat the test from PC1 if required.
10. Use Simulation mode to observe ARP and ICMP packets, then save the Packet Tracer project.

IP ADDRESS CONFIGURATION:

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet0	192.168.10.10	255.255.255.0	192.168.10.1
PC1	FastEthernet0	192.168.20.10	255.255.255.0	192.168.20.1
Router	G0/0	192.168.10.1	255.255.255.0	—
Router	G0/1	192.168.20.1	255.255.255.0	—

OUTPUT:

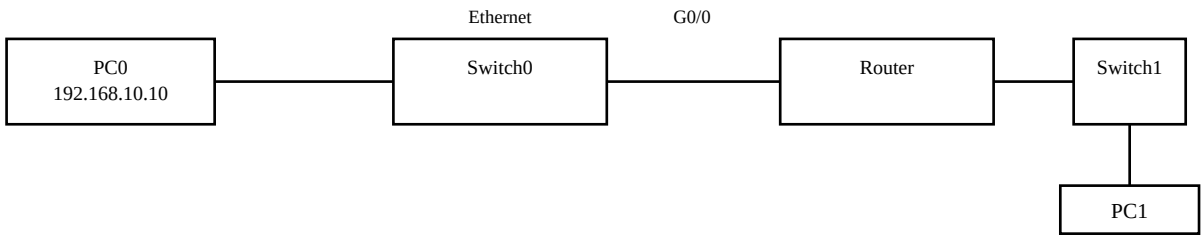

Fig. 1. Example internetwork topology. Connect the second switch to the router's second LAN interface; interface names may vary by model.

ROUTER CONFIGURATION:

```
Router> enable
```

```
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# end
Router# show ip interface brief
```

OUTPUT:

Sample router interface status:

```
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
GigabitEthernet0/1     192.168.20.1    YES manual up                    up
```

PC0 CONNECTIVITY TEST:

```
C:\> ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics:
    Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\> ping 192.168.20.10

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
```

OBSERVATION:

1. The Ethernet link indicators become active when cables are connected to valid interfaces and the interfaces are enabled.

2. Each PC communicates with its local default gateway when the IP address and subnet mask are correct.

3. The router forwards packets between the two different IP subnets.

4. Successful ping replies from PC1 confirm end-to-end connectivity between the two networks.

5. Simulation mode can show ARP resolution on the local segment and ICMP echo request/reply traffic routed between subnets.

RESULT:

Thus, an internetwork was designed and configured using switches, a router and Ethernet cables. IP addresses and router interfaces were configured, and connectivity between devices on different subnets was tested using ping.

Note: Outputs shown are illustrative sample outputs. Replace or supplement them with screenshots from your actual Packet Tracer or laboratory setup if required.